

Time primitives & definitions:

N = parameter count, C = compute (FLOP/s), β = bandwidth (bytes/s)

L_{in} = input token count L_{out} = output token count $L_{avg} = L_{in} + L_{out}/2$

$$t_{\text{weight}} = \frac{Nb_w}{\beta} \quad t_{kv}(L) = \frac{\kappa L}{\beta} \quad (\text{per sequence}) \quad t_c = \frac{2N}{C} \quad (\text{per token})$$

where multiply-accumulate is 2 FLOPs, and

$$\kappa = 2 \cdot \underbrace{n_{\text{layers}}}_{\text{transformer layers}} \cdot \underbrace{n_{kv}}_{\text{KV-heads}} \cdot \underbrace{d_h}_{\text{head-dim}} \cdot \underbrace{b_{kv}}_{\text{bytes per KV element}} \quad b_w = \text{bytes per weight}$$

$$t_{\text{attn}}(L) = aL, \quad a = \frac{4 n_{\text{layers}} n_q d_h}{C} \quad t_{\text{attn},d}(L) = a_d L, \quad a_d = \frac{4 n_{\text{layers},d} n_{q,d} d_{h,d}}{C}$$

Prefill attends over all L_{in} positions at once. Causal masking leaves $\sum_{i=1}^{L_{in}} i = L_{in}(L_{in} + 1)/2$ query-context pairs, so per sequence:

$$T_{\text{attn}}^{\text{pre}}(L) = \frac{1}{2} aL(L + 1) \approx \frac{1}{2} aL^2, \quad T_{\text{attn},d}^{\text{pre}}(L) = \frac{1}{2} a_d L^2$$

Prefill cost:

$$T_{\text{prefill}}^{\text{AR}} = \max(B[t_c L_{in} + \frac{1}{2} a L_{in}^2](1 - h), t_w) + B t_{kv}(L_{in})(1 - h) \quad (h = \text{prefix cache hit rate})$$

$$T_{\text{prefill}}^{\text{spec}} = T_{\text{prefill}}^{\text{AR}} + [\max(B[t_{c,d} L_{in} + \frac{1}{2} a_d L_{in}^2](1 - h), t_{w,d}) + B t_{kv,d}(L_{in})(1 - h)] \quad (d = \text{drafter})$$

Decode pass cost:

$$T_{\text{AR}}(L) = \max(B(t_c + t_{\text{attn}}(L)), t_w) + B t_{kv}(L + 1) \quad (\text{no KV overlap when compute bound})$$

$$T_{\text{spec}}(L) = T_{\text{spec}}^{\text{tgt}}(L) + T_{\text{spec}}^{\text{draft}}(L)$$

$$T_{\text{spec}}^{\text{tgt}}(L) = \max((V + 1)B(t_c + t_{\text{attn}}(L)), t_w) + B t_{kv}(L + V + 1)$$

$$T_{\text{spec}}^{\text{draft}}(L) = D [\max(B(t_{c,d} + t_{\text{attn},d}(L)), t_{w,d}) + B t_{kv,d}(L)]$$

End-to-end speedup:

$$T_{\text{decode}}^{\text{AR}} = L_{out} \cdot T_{\text{AR}}(L_{avg})$$

$$T_{\text{decode}}^{\text{spec}} = \frac{L_{out}}{\Omega} \cdot T_{\text{spec}}(L_{avg})$$

$$S_{e2e} = \frac{T_{\text{prefill}}^{\text{AR}} + T_{\text{decode}}^{\text{AR}}}{T_{\text{prefill}}^{\text{spec}} + T_{\text{decode}}^{\text{spec}}}$$

Achievable batch:

$$B_{\text{ach}} = \min(B_{KV}, B_{\text{conc}})$$

$$B_{KV} = \frac{TP \cdot (M_{\text{HBM}} - M_{\text{reserve}}) - Nb_w - N_d b_{w,d}}{(\kappa_{\text{eff}} + \kappa_{d,\text{eff}}) \cdot L_{\text{in+out}}} \quad \left(\kappa_{\text{eff}} = \kappa \cdot \max\left(1, \frac{TP}{n_{kv}}\right) \right)$$

Assumptions:

Each is listed with the direction and, where measured, the size of its error (see `MODEL_GAPS.md`; Llama-3.1-8B + EAGLE-3 on A100-40).

1. **Weight streaming and compute overlap perfectly:** the $\max(\cdot, \cdot)$ charges only the larger and treats the smaller as free. *Optimistic.* Where both passes are weight bound the model predicts a verify pass costs the same as an AR pass (ratio 1.00); measured 1.35–1.46 at $B \leq 32$. This is the dominant error.
2. **Peak bandwidth and FLOP/s are achieved.** *Optimistic.* The AR forward alone runs 1.18–1.35 $\times$ the weight-bound prediction at one query token per sequence. Largely cancels in a ratio.
3. **Tensor-parallel communication is free.** S is TP invariant here; TP enters only through B_{KV} . Unmeasured; false on PCIe parts.
4. **Per-step overhead is zero:** scheduling, sampling, rejection bookkeeping, block tables. Measured at 3–5% of a step on both arms, so this one is nearly true.
5. **Ω is independent of B and L_{in} ,** depending only on L_{out} . Measured true across the Phase 1 grid (2.24–2.70, set by L_{out} alone).
6. $\kappa_d = \kappa/n_{\text{layers}}$: the drafter reads one decoder layer of KV. *Optimistic:* measured 1.6 $\times$ low.
7. **No KV/compute overlap when compute bound** (the additive $B t_{kv}$ term), and prefill is compute bound at the batches considered.
8. **Greedy decoding**, so Ω is deterministic, and a chain drafter with $D = V$.